

Honeypot Deployment and Monitoring

Final Report

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1. Introduction

- **Objective:** This project focuses on deploying a honeypot to simulate and monitor malicious activities in a controlled environment. A honeypot was configured to mimic a vulnerable system and capture attacker behavior such as login attempts, credential usage, and command execution.
- The project aims to understand real-world attack patterns and analyze how attackers interact with exposed services.

2. Lab Setup

- A controlled lab environment was established using virtualization to simulate real-world attack and defense scenarios. Two virtual machines were configured on the same network to enable communication between the attacker and the target system.
 - **Attacker Machine:** Parrot Security OS
 - **Target Machine:** Ubuntu Server
 - **Honeypot Tool:** Cowrie
- Both systems were connected using an Internal (host-only) network configuration to ensure isolated and secure communication within the lab.
- Connectivity between the machines was verified using ICMP (ping), confirming successful network communication.

```
pbantu@bantu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:23:96:54 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe23:9654/64 scope link
        valid_lft forever preferred_lft forever
pbantu@bantu:~$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=1.85 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.50 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=2.12 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=64 time=2.32 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=64 time=1.83 ms
^C
--- 192.168.56.102 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4010ms
rtt min/avg/max/mdev = 1.503/1.924/2.322/0.278 ms
pbantu@bantu:~$ _
```

```
[ceaser@parrot]~$ ifconfig
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:3e:a3:02 brd ff:ff:ff:ff:ff:ff
    altname enx0800273ea302
    inet 192.168.56.102/24 brd 192.168.56.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe3e:a302/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[ceaser@parrot]~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.13 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.51 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=1.73 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=1.79 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=1.48 ms
^C
--- 192.168.56.101 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4040ms
rtt min/avg/max/mdev = 1.134/1.526/1.792/0.231 ms
[ceaser@parrot]~$
```

- The screenshot shows successful communication between the attacker and target machines, confirming proper network configuration.
- Both systems are reachable within the same network, enabling further testing.

3. Methodology

- The project was carried out in the following phases:
 - **Reconnaissance:** Identifying live hosts in the network
 - **Scanning:** Discovering open ports and services
 - **Enumeration:** Gathering detailed service information
 - **Exploitation:** Simulating attacks on the target
 - **Post-Exploitation:** Observing attacker activity and analyzing logs

4. Implementation

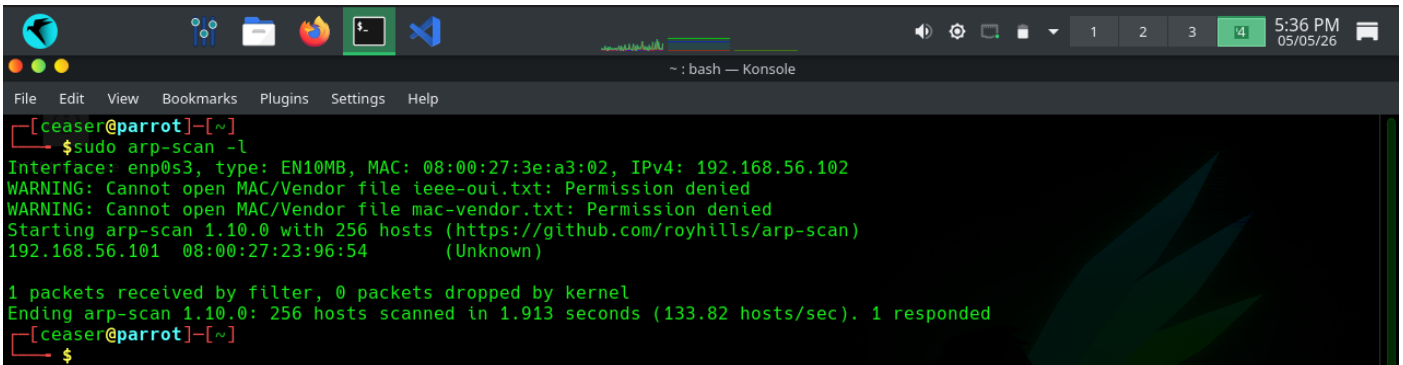
- **Honeytrap Deployment (Cowrie) – Ubuntu Server**
 - The Cowrie honeypot was deployed on the target system to mimic a vulnerable SSH service and capture malicious activities.

```
pbantu@bantu:~$ sudo su - cowrie
cowrie@bantu:~$ cd cowrie/
cowrie@bantu:~/cowrie$ source cowrie-env/bin/activate
(cowrie-env) cowrie@bantu:~/cowrie$ cowrie start
Starting cowrie: [twistd --umask=0022 --pidfile /home/cowrie/cowrie/var/run/cowrie.pid --logger cowrie.python.logfile.logger cowrie]...
/home/cowrie/cowrie/cowrie-env/lib/python3.10/site-packages/twisted/conch/ssh/transport.py:110: CryptographyDeprecationWarning: TripleDES has been moved to cryptography.hazmat.decrepit.ciphers.algorithms.TripleDES and will be removed from cryptography.hazmat.primitives.ciphers.algorithms in 48.0.0.
  b"3des-cbc": (algorithms.TripleDES, 24, modes.CBC),
/home/cowrie/cowrie/cowrie-env/lib/python3.10/site-packages/twisted/conch/ssh/transport.py:117: CryptographyDeprecationWarning: TripleDES has been moved to cryptography.hazmat.decrepit.ciphers.algorithms.TripleDES and will be removed from cryptography.hazmat.primitives.ciphers.algorithms in 48.0.0.
  b"3des-ctr": (algorithms.TripleDES, 24, modes.CTR),
(cowrie-env) cowrie@bantu:~/cowrie$
```

- The honeypot service is actively running and ready to capture malicious activity.
- The system is configured to simulate a vulnerable SSH environment.

▪ Reconnaissance – ARP Scan

- A network discovery scan was performed to identify active hosts within the local network using ARP requests.



```

[ceaser@parrot]~$ sudo arp-scan -l
Interface: enp0s3, type: EN10MB, MAC: 08:00:27:3e:a3:02, IPv4: 192.168.56.102
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.56.101 08:00:27:23:96:54 (Unknown)

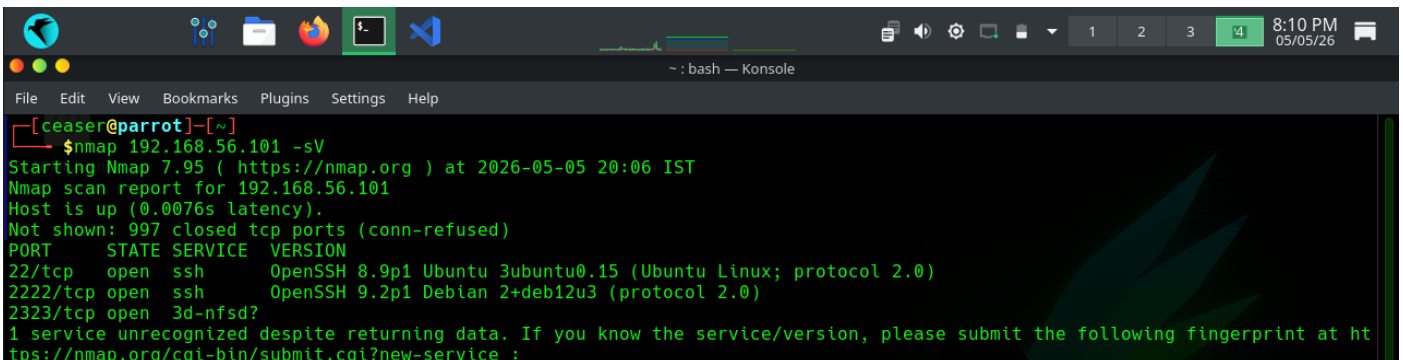
1 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.913 seconds (133.82 hosts/sec). 1 responded
[ceaser@parrot]~$

```

- The scan output displays active hosts in the network along with their IP and MAC addresses.
- The target system was successfully identified among the live hosts.

▪ Scanning – Nmap Scan

- A port scan was conducted to detect open ports, running services, and their versions on the target system.



```

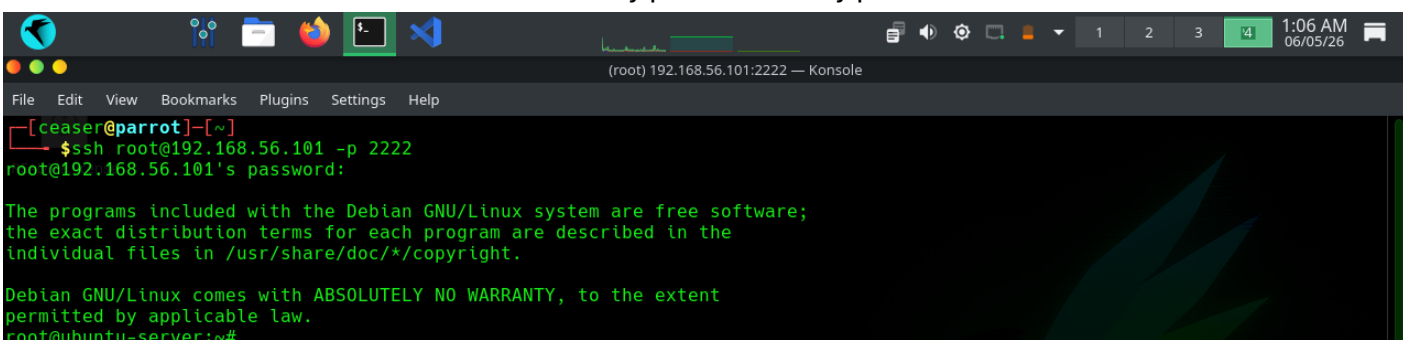
[ceaser@parrot]~$ nmap 192.168.56.101 -sV
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-05 20:06 IST
Nmap scan report for 192.168.56.101
Host is up (0.0076s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.15 (Ubuntu Linux; protocol 2.0)
2222/tcp  open  ssh      OpenSSH 9.2p1 Debian 2+deb12u3 (protocol 2.0)
2323/tcp  open  3d-nfsd?
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :

```

- The scan results reveal open ports and corresponding services running on the target system.
- Service version detection provides valuable information for further analysis.

▪ Enumeration – Service Interaction

- Further enumeration was performed by interacting with the identified services to gather detailed information and identify potential entry points and weaknesses.



```

[ceaser@parrot]~$ ssh root@192.168.56.101 -p 2222
root@192.168.56.101's password:

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@ubuntu-server:~#

```

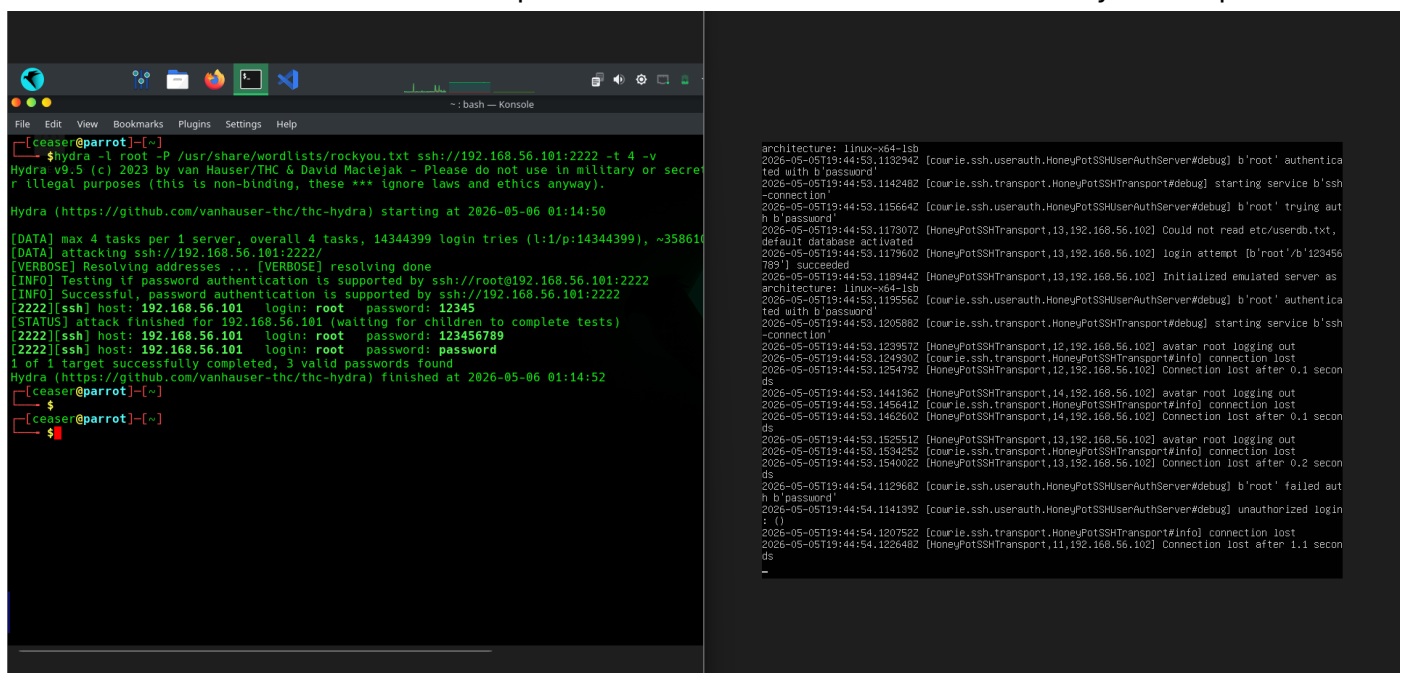
- The screenshot demonstrates interaction with identified service to gather additional information.

```

2026-05-05T19:35:57.606855Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' authenticated with b'password'
2026-05-05T19:35:57.607495Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-connection'
2026-05-05T19:35:57.610805Z [cowrie.ssh.connection.CowrieSSHConnection#debug] got channel b'session' request
2026-05-05T19:35:57.611336Z [cowrie.ssh.session.HoneyPotSSHSession#info] channel open
2026-05-05T19:35:57.611523Z [cowrie.ssh.connection.CowrieSSHConnection#debug] got global b'no-more-sessions@openssh.com' request
2026-05-05T19:35:57.727161Z [twisted.conch.ssh.session#info] Handling pty request: b'xterm-256color' (36, 125, 1251, 700)
2026-05-05T19:35:57.728421Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] Terminal Size: 125 36
2026-05-05T19:35:57.729563Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] request_env: COLORTERM=truecolor
2026-05-05T19:35:57.730461Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] request_env: LC_ADDRESS=en_IN
2026-05-05T19:35:57.731280Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] request_env: LC_NAME=en_IN
2026-05-05T19:35:57.732113Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] request_env: LC_MONETARY=en_IN
2026-05-05T19:35:57.733409Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] request_env: LC_PAPER=en_IN
2026-05-05T19:35:57.734437Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] request_env: LANG=en_IN
2026-05-05T19:35:57.735382Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] request_env: LC_IDENTIFICATION=en_IN
2026-05-05T19:35:57.736515Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] request_env: LC_TELEPHONE=en_IN
2026-05-05T19:35:57.738156Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] request_env: LC_MEASUREMENT=en_IN
2026-05-05T19:35:57.739092Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] request_env: LC_TIME=en_IN
2026-05-05T19:35:57.739923Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,4,192.168.56.102] request_env: LC_NUMERIC=en_IN
2026-05-05T19:35:57.740806Z [twisted.conch.ssh.session#info] Getting shell
^[_S_

```

- This step helps in identifying potential entry points and weaknesses.
- **Exploitation – Brute Force SSH using Hydra**
 - Attack simulation was performed from the attacker machine using SSH login attempts and brute-force techniques to mimic real-world attacks and test system exposure.



The image shows two side-by-side screenshots. The left screenshot is a terminal window titled 'bash - Konsole' showing the execution of Hydra. The user is at the prompt 'ceaser@parrot' and runs the command: `hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.101:2222 -t 4 -v`. The output shows that Hydra successfully found three passwords for the root user on the target IP 192.168.56.101: 12345, 123456789, and password. The right screenshot is a log file showing the detailed interaction between the attacker's Hydra and the target's Cowrie SSH service. It shows the initial connection attempt, the successful authentication of the root user with the password 'password', and the subsequent connection loss after 0.1 seconds.

```

ceaser@parrot:~$ hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.101:2222 -t 4 -v
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret
r illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-06 01:14:50

[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l1/p:14344399), ~35861
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://root@192.168.56.101:2222
[INFO] Successful, password authentication is supported by ssh://192.168.56.101:2222
[2222][ssh] host: 192.168.56.101 login: root password: 12345
[STATUS] attack finished for 192.168.56.101 (waiting for children to complete tests)
[2222][ssh] host: 192.168.56.101 login: root password: 123456789
[2222][ssh] host: 192.168.56.101 login: root password: password
1 of 1 target successfully completed, 3 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-06 01:14:52
ceaser@parrot:~$
ceaser@parrot:~$

```

```

architecture: linux-x64-1sb
2026-05-05T19:44:53.113294Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' authenticated with b'password'
2026-05-05T19:44:53.114248Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-connection'
2026-05-05T19:44:53.115664Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' trying authentication with b'password'
2026-05-05T19:44:53.117307Z [HoneyPotSSHTransport,13,192.168.56.102] Could not read etc/userdb.txt, default database activated.
2026-05-05T19:44:53.117602Z [HoneyPotSSHTransport,13,192.168.56.102] login attempt [b'root'/b'123456789'] succeeded
2026-05-05T19:44:53.118944Z [HoneyPotSSHTransport,13,192.168.56.102] Initialized emulated server as architecture: linux-x64-1sb
2026-05-05T19:44:53.119556Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' authenticated with b'password'
2026-05-05T19:44:53.120588Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-connection'
2026-05-05T19:44:53.123957Z [HoneyPotSSHTransport,12,192.168.56.102] avatar root logging out
2026-05-05T19:44:53.124930Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-05-05T19:44:53.125479Z [HoneyPotSSHTransport,12,192.168.56.102] connection lost after 0.1 seconds
2026-05-05T19:44:53.144136Z [HoneyPotSSHTransport,14,192.168.56.102] avatar root logging out
2026-05-05T19:44:53.145641Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-05-05T19:44:53.146260Z [HoneyPotSSHTransport,14,192.168.56.102] connection lost after 0.1 seconds
2026-05-05T19:44:53.152551Z [HoneyPotSSHTransport,13,192.168.56.102] avatar root logging out
2026-05-05T19:44:53.153425Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-05-05T19:44:53.154002Z [HoneyPotSSHTransport,13,192.168.56.102] connection lost after 0.2 seconds
2026-05-05T19:44:54.112968Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' failed authentication with b'password'
2026-05-05T19:44:54.114139Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] unauthorized login attempt
2026-05-05T19:44:54.120752Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-05-05T19:44:54.122648Z [HoneyPotSSHTransport,11,192.168.56.102] Connection lost after 1.1 seconds

```

- The screenshot shows simulated attack attempts using brute-force techniques against the target system.
- Multiple login attempts indicate testing of weak credentials.

▪ Post-Exploitation – Monitoring and Logging

- The honeypot successfully recorded:
 - Login attempts
 - Username and password combinations
 - Commands executed by the attacker

```
ansport,4,192.168.56.102] request_env: LC_IDENTIFICATION=en_IN
2026-05-05T19:35:57.736515Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTr
ansport,4,192.168.56.102] request_env: LC_TELEPHONE=en_IN
2026-05-05T19:35:57.738156Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTr
ansport,4,192.168.56.102] request_env: LC_MEASUREMENT=en_IN
2026-05-05T19:35:57.739092Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTr
ansport,4,192.168.56.102] request_env: LC_TIME=en_IN
2026-05-05T19:35:57.739923Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTr
ansport,4,192.168.56.102] request_env: LC_NUMERIC=en_IN
2026-05-05T19:35:57.740806Z [twisted.conch.ssh.session#info] Getting shell
^[S[28215.614074] watchdog: BUG: soft lockup - CPU#1 stuck for 28s! [systemd-resolve:1921]
^[S[2026-05-05T19:38:20.556181Z [HoneyPotSSHTransport,4,192.168.56.102] CMD: whoami
2026-05-05T19:38:20.563352Z [HoneyPotSSHTransport,4,192.168.56.102] Command found: whoami
2026-05-05T19:38:29.589264Z [HoneyPotSSHTransport,4,192.168.56.102] CMD: uname -a
2026-05-05T19:38:29.592385Z [HoneyPotSSHTransport,4,192.168.56.102] Command found: uname -a
2026-05-05T19:38:38.929390Z [HoneyPotSSHTransport,4,192.168.56.102] CMD: pwd
2026-05-05T19:38:38.930509Z [HoneyPotSSHTransport,4,192.168.56.102] Command found: pwd
2026-05-05T19:38:42.230390Z [HoneyPotSSHTransport,4,192.168.56.102] CMD: ls
2026-05-05T19:38:42.231563Z [HoneyPotSSHTransport,4,192.168.56.102] Command found: ls
2026-05-05T19:38:52.249175Z [HoneyPotSSHTransport,4,192.168.56.102] CMD: ls -a
2026-05-05T19:38:52.250673Z [HoneyPotSSHTransport,4,192.168.56.102] Command found: ls -a
2026-05-05T19:39:02.307620Z [HoneyPotSSHTransport,4,192.168.56.102] CMD: exit
2026-05-05T19:39:02.309189Z [HoneyPotSSHTransport,4,192.168.56.102] Command found: exit
2026-05-05T19:39:02.310120Z [twisted.conch.ssh.session#info] exitCode: 0
2026-05-05T19:39:02.310635Z [cowrie.ssh.connection.CowrieSSHConnection#debug] sending request b'exit
-status'
2026-05-05T19:39:02.311474Z [HoneyPotSSHTransport,4,192.168.56.102] Closing TTY Log: var/lib/cowrie/
tty/c435baab819b888049b70f72de4daf5212f2b124a4b307b85b1e5d88479e1f24 after 184.6 seconds
2026-05-05T19:39:02.313285Z [cowrie.ssh.connection.CowrieSSHConnection#info] sending close 0
2026-05-05T19:39:02.317111Z [cowrie.ssh.session.HoneyPotSSHSession#info] remote close
2026-05-05T19:39:02.319787Z [HoneyPotSSHTransport,4,192.168.56.102] Got remote error, code 11 reason
: b'disconnected by user'
2026-05-05T19:39:02.322916Z [HoneyPotSSHTransport,4,192.168.56.102] avatar root logging out
2026-05-05T19:39:02.324427Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-05-05T19:39:02.324982Z [HoneyPotSSHTransport,4,192.168.56.102] Connection lost after 200.7 seco
nds
-
```

- The attacker executed basic system commands, indicating an attempt to explore the compromised system.
- The honeypot logs were monitored to capture attacker behavior, including login attempts, credentials used, and commands executed.
- The captured logs provide insight into attacker techniques and commonly used credentials.
- Common usernames such as *root* were frequently targeted
- Weak passwords like '123456' were used in brute-force attempts
- These findings highlight typical attacker behavior during initial system compromise.

```

default database activated
2026-05-06T08:02:37.288085Z [HoneyPotSSHTransport,3,192.168.56.102] login attempt [b'root'/b'123456789'] succeeded
2026-05-06T08:02:37.289473Z [HoneyPotSSHTransport,3,192.168.56.102] Initialized emulated server as a
rchitecture: linux-x64-lsb
2026-05-06T08:02:37.290322Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' authentica
ted with b'password'
2026-05-06T08:02:37.291662Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh
-connection'
2026-05-06T08:02:37.293461Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' trying aut
h b'none'
2026-05-06T08:02:37.296663Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' trying aut
h b'password'
2026-05-06T08:02:37.297224Z [HoneyPotSSHTransport,1,192.168.56.102] Could not read etc/userdb.txt, d
efault database activated
2026-05-06T08:02:37.297426Z [HoneyPotSSHTransport,1,192.168.56.102] login attempt [b'root'/b'1234567
89'] failed
2026-05-06T08:02:37.313567Z [HoneyPotSSHTransport,4,192.168.56.102] avatar root logging out
2026-05-06T08:02:37.314075Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-05-06T08:02:37.314225Z [HoneyPotSSHTransport,4,192.168.56.102] Connection lost after 0.1 second
s
2026-05-06T08:02:37.319517Z [HoneyPotSSHTransport,3,192.168.56.102] avatar root logging out
2026-05-06T08:02:37.319906Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-05-06T08:02:37.320167Z [HoneyPotSSHTransport,3,192.168.56.102] Connection lost after 0.1 second
s
2026-05-06T08:02:37.323152Z [HoneyPotSSHTransport,2,192.168.56.102] avatar root logging out
2026-05-06T08:02:37.323690Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-05-06T08:02:37.324099Z [HoneyPotSSHTransport,2,192.168.56.102] Connection lost after 0.1 second
s
2026-05-06T08:02:38.300268Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' failed aut
h b'password'
2026-05-06T08:02:38.301258Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] unauthorized login
: ()
2026-05-06T08:02:38.307455Z [cowrie.ssh.transport.HoneyPotSSHTransport#info] connection lost
2026-05-06T08:02:38.307957Z [HoneyPotSSHTransport,1,192.168.56.102] Connection lost after 1.1 second
s

```

- The log output captures unauthorized login attempts along with credentials used. The attacker executed basic system commands such as `'ls'` and `'whoami'` indicate an attempt to explore and enumerate the compromised system. The source IP address is also successfully recorded.

5. Conclusion

- The project successfully demonstrated the deployment of a honeypot and the simulation of attacks in a controlled environment. The captured data provided valuable insights into attacker techniques, emphasizing the importance of monitoring and securing exposed services.

6. Recommendations

- **Long-Term Measures**
 - Implement firewall rules to restrict unauthorized access
 - Deploy intrusion detection/prevention systems
 - Conduct regular vulnerability assessments
- **Immediate Fixes**
 - Remove or disable vulnerable services
 - Patch outdated services

7. Key Learnings

- Gained hands-on experience in deploying and configuring a honeypot
- Understood attacker behavior and common attack patterns
- Learned how to monitor and analyze security logs
- Improved practical knowledge of network security and attack simulation

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